

Question 1. What is a conditional statement in programming?

Answer: A conditional statement helps the program make decisions based on whether a condition is true or false.

Question 2. What does an if statement do?

Answer: It executes a block of code only when a specified condition is true.

Question 3. What is the purpose of an else block?

Answer: The else block runs when the condition in the if statement is false.

Question 4. When is an else block executed?

Answer: It is executed when the condition in the if statement is not satisfied.

Question 5. What is a Boolean expression?

Answer: It is an expression that gives either true or false as the result.

Question 6. Write the syntax of a simple if statement.

Answer:

```
if(condition)
{
    // statements
}
```

Question 7. What operator is commonly used to test equality?

Answer: The == operator is used to compare two values for equality.

Question 8. What does the modulus (%) operator return?

Answer: It returns the remainder after division.

Question 9. How can you check if a number is even?

Answer: By checking if `number % 2 == 0`.

Question 10. How can you check if a number is odd?

Answer: By checking if `number % 2 != 0`.

Question 11. What is the purpose of else-if?

Answer: It is used to check multiple conditions one after another.

Question 12. What is a nested if statement?

Answer: A nested if statement is an if statement placed inside another if statement.

Question 13. Can an if statement exist without an else?

Answer: Yes, an if statement can be used without an else block.

Question 14. What is the output when a false condition is tested in an if statement without else?

Answer: Nothing happens because the code inside the if block is skipped.

Question 15. What is the role of braces {} in C/C++ conditionals?

Answer: Braces are used to group multiple statements together.

Question 16. What does > mean?

Answer: It means "greater than".

Question 17. What does < mean?

Answer: It means "less than".

Question 18. What does >= mean?

Answer: It means "greater than or equal to".

Question 19. What does <= mean?

Answer: It means "less than or equal to".

Question 20. What does != mean?

Answer: It means "not equal to".

Question 21. What is a switch statement?

Answer: A switch statement is used to select one option from many possible choices.

Question 22. What is a case label in a switch statement?

Answer: A case label represents a particular value that the switch expression is compared with.

Question 23. Why is break used in switch?

Answer: It stops the execution after the matching case is completed.

Question 24. What happens if break is omitted?

Answer: The program continues to execute the next cases, which is called fall-through.

Question 25. What is the default case in switch?

Answer: It is executed when none of the case labels match.

Question 26. Can switch compare strings in C++?

Answer: No, switch statements generally work with integer or character values, not strings.

Question 27. Which statement is better for many constant choices: if-else or switch?

Answer: Switch is usually better because it makes the code easier to read.

Question 28. How do you test whether a value is positive?

Answer: By checking if the value is greater than zero.

```
num > 0
```

Question 29. How do you test whether a value is negative?

Answer: By checking if the value is less than zero.

```
num < 0
```

Question 30. How do you test whether a value is zero?

Answer: By checking if the value is equal to zero.

```
num == 0
```

Question 31. Write a condition to check if age is at least 18.

Answer:

`age >= 18`

Question 32. Write a condition to check if marks are above 50.

Answer:

`marks > 50`

Question 33. What is decision making in programming?

Answer: It is the process of choosing actions based on conditions.

Question 34. Can nested if statements have multiple levels?

Answer: Yes, nested if statements can have many levels.

Question 35. What is logical AND (&)?

Answer: It returns true only if both conditions are true.

Question 36. What is logical OR (||)?

Answer: It returns true if at least one condition is true.

Question 37. What is logical NOT (!)?

Answer: It reverses the result of a condition.

Question 38. How do you check if a year is divisible by 4?

Answer:

`year % 4 == 0`

Question 39. What is a leap year?

Answer: A leap year is a year having 366 days instead of 365 days.

Question 40. What is the first condition for a leap year?

Answer: The year should be divisible by 4.

Question 41. How do you find the largest of two numbers?

Answer: Compare the two numbers and select the greater one.

Question 42. How do you find the largest of three numbers?

Answer: By comparing all three numbers using if-else statements.

Question 43. What is an if-else-if ladder?

Answer: It is a sequence of conditions checked one after another until one becomes true.

Question 44. What is the difference between if and switch?

Answer: The if statement can handle complex conditions, whereas switch is used for fixed constant values.

Question 45. Can switch use ranges directly?

Answer: No, switch cannot directly check ranges.

Question 46. What is fall-through in switch?

Answer: It is the execution of subsequent cases when break is not used.

Question 47. What data type is usually used for conditions?

Answer: The bool data type is commonly used.

Question 48. Why are conditional statements important?

Answer: They allow programs to make decisions and perform different actions according to different situations.

Question 49. Give one real-life example of a conditional statement.

Answer: If the traffic light is green, vehicles move; otherwise, they stop.

Question 50. Name four conditional structures covered.

Answer:

1. if statement
2. if-else statement
3. else-if ladder
4. switch statement